

LM06 Fixed-Income Bond Valuation - Prices and Yields

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1. Introduction

This learning module covers:

- How to price a bond on or between coupon dates
- Relationship between a bond's price and bond features such as coupon rate, maturity, and yield-to-maturity
- Matrix pricing

2. Bond Pricing and The Time Value of Money

Bond Pricing with a Market Discount Rate

A bond's price is the present value of all future cash flows at the market discount rate. The discount rate is the rate of return required by investors given the risk of investment in the bond. It is also known as the **required yield**, or **required rate of return**.

$$\text{PV of bond} = \frac{\text{PMT}}{(1+r)^1} + \frac{\text{PMT}}{(1+r)^2} + \cdots + \frac{\text{PMT}+\text{FV}}{(1+r)^N}$$

where:

PMT = coupon payment per period

FV = par value of the bond paid at maturity

r = market discount rate

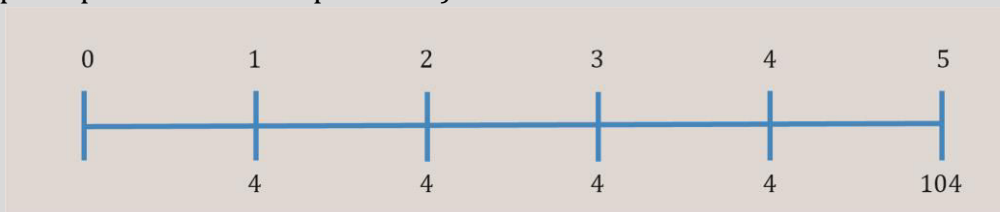
N = number of periods until maturity

Example

The coupon rate on a bond is 4% and the payment is made once a year. The time to maturity is five years and the market discount rate is 6%. What is the bond price per 100 of par value?

Solution:

Start by drawing a timeline for the cash flows. The par value of the bond or principal is \$100. A coupon payment of \$4 is made every year. At maturity (at the end of five years), a payment of \$104 (principal of \$100 + coupon of \$4) is made.



We are required to calculate the present value of bond at time $t = 0$. For that, we discount all the future cash flows at the market discount rate of 6%.

End of year	Type of cash flow	Amount	Present value
1	Coupon	4	3.77
2	Coupon	4	3.55

3	Coupon	4	3.36
4	Coupon	4	3.17
5	Coupon + principal	104	77.71
			91.56

Using the formula, $PV = \frac{4}{1.06} + \frac{4}{1.06^2} + \frac{4}{1.06^3} + \frac{4}{1.06^4} + \frac{104}{1.06^5} = 91.575$

Note: The table and formula above were just for your understanding.

On the exam, solve this quickly using a financial calculator:

N = 5; I/Y = 6; FV = 100; PMT = 4; PV = ? PV = -91.575

Notice the present value of the bond (91.575) is lower than the par value of the bond (100).

Premium, Par, and Discount Bonds

Example

Assume we have another bond with a coupon rate of 8%, paid annually. If the market discount rate is again 6%, what is the price of the bond?

Solution:

Note that the coupon rate here is higher than the market rate. Calculate PV using these input values: N = 5; I/Y = 6; PMT = 8; FV = 100; CPT PV

PV = -108.425. In this case, the present value of the bond is higher than its par value. Such a bond is called a premium bond. Since the coupon rate (8%) is higher than the required return of 6%, investors pay more than the par value for the bond.

What if the coupon rate is 6%?

N = 5; I/Y = 6; PMT = 6; FV = 100; CPT PV

PV = -100. The present value is equal to the par value of the bond.

What if the coupon rate is 2%?

N = 5; I/Y = 6; PMT = 2; FV = 100; CPT PV

PV = -83.151. This type of bond, which sells at a price below its par value, is called a discount bond. Investors require a return of 6% but it pays a coupon of only 2%, so investors pay less.

Par, premium, or discount	Relationship between coupon rate & market discount rate
Par bond	Coupon rate = market discount rate
Discount bond	Coupon rate < market discount rate
Premium bond	Coupon rate > market discount rate

Yield-to-Maturity

The yield to maturity (YTM) is the internal rate of return on the cash flows – the uniform interest rate that will make the sum of the present values of future cash flows equal to the

price of the bond. It is the implied market discount rate. In simpler terms, it is a bond's internal rate of return - the rate of return on a bond including interest payments and capital gain if the bond is held until maturity. Yield to maturity is based on three important assumptions:

- The investor holds the bond to maturity.
- The issuer does not default on payments and pays coupon and principal as they come due.
- The investor is able to reinvest all proceeds (coupons) at the YTM. (This is an unrealistic assumption as the interest rates may increase or decrease after the bond is purchased. So, the coupon may be reinvested at a higher or lower rate.)

Example

A \$100 face value bond with a coupon rate of 10% has a maturity of 4 years. The price of the bond is \$80. What is its yield to maturity?

Solution:

Based on YTM's definition, we come up with the equation below:

$$80 = \frac{10}{(1+r)^1} + \frac{10}{(1+r)^2} + \frac{10}{(1+r)^3} + \frac{110}{(1+r)^4}$$

This equation can be solved using trial and error but it is much easier to use a financial calculator: N = 4; PMT = 10; FV = 100; PV = -80; CPT I/Y = 17.34%. The bond is trading at a discount. So, the coupon rate (10%) must be lower than market rate (17.34%).

Example

Calculate the yields to maturity for the following bonds:

Bond	Coupon Payment per Period	Number of Periods to Maturity	Price
A	2.50	3	102.80
B	2.00	5	97.76
C	0.00	48	23.425

Solution:

$$\text{Bond A: } 102.80 = \frac{2.5}{(1+r)} + \frac{2.5}{(1+r)^2} + \frac{102.5}{(1+r)^3}, r = 1.54\%$$

$$\text{Bond B: } 97.76 = \frac{2}{(1+r)} + \frac{2}{(1+r)^2} + \frac{2}{(1+r)^3} + \frac{2}{(1+r)^4} + \frac{102}{(1+r)^5}, r = 2.48\%$$

$$\text{Bond C: } 23.425 = \frac{100}{(1+r)^{48}}, r = 3.07\%$$

Flat Price, Accrued Interest, and the Full Price

When a bond is between coupon dates, its price has two parts: flat price and accrued interest. The sum of the two parts is called the full price or dirty price.

Full price or dirty price = Flat price + Accrued interest

$$PV^{\text{full}} = PV^{\text{flat}} + AI$$

Flat price = Full price – Accrued interest

$$\text{Accrued interest} = \frac{t}{T} \times PMT$$

where:

t = number of days since the last coupon payment

T = number of days between coupon payments

PMT = interest payment per period

As indicated in the above equation, the accrued interest is based on the number of days since the last coupon payment and the number of days between coupon payments. Here are two conventions for calculating the number of days:

- Actual convention: Uses the actual number of days, including weekends, holidays and leap days.
- 30/360 convention: Assumes 30 days in a month and 360 days in a year.

Illustration of Flat Price and Dirty Price through an Example

Assume you own a two-year bond issued on 10 January 2017 with a par value of \$100. The bond pays a coupon of 8% half-yearly. The first coupon is due on 10 July 2017, the next on 10 January 2018, the third on 10 July 2018, and the final one is paid along with principal at maturity date on 10 January 2019. The market discount rate is equal to the coupon rate and so the flat price is 100. An investor wants to purchase the bond from you on 31 May 2017. What is the full price he must pay?

You have held the bond for 140 days (since the bond was issued) and earned an interest for this period even though it has not been paid. The next coupon payment is \$4. Assuming each period is 180 days the accrued interest is $4 \times (140/180) = 3.11$. The full price is $100.00 + 3.11 = 103.11$. This is the amount you will receive on selling the bond.

Note that it is the flat (or clean) price which is quoted. The accrued interest increases every day until the coupon payment. So, if the accrued interest was included there would be a variation in bond quotes on a daily basis and a sudden drop after the coupon payment is made, which is misleading.

The full price of a bond can also be calculated using the equation shown below.

$$PV_{\text{full}} = PV \times (1 + r)^{\frac{t}{T}}$$

where:

t = number of days since the last coupon payment

T = number of days between coupon payments

PV = present value of the bond (not the flat price)

Example

A 5% French corporate bond is priced for settlement on 14 July 2015. The bond makes semi-annual coupon payments on 1 March and 1 September of each year and matures on 1 September 2020. Assuming a 30/360 day count convention for accrued interest, calculate the full price, the accrued interest, and the flat price per EUR 100 of par value for the following three yields to maturity: (I) 4.75%, (II) 5.00% and (III) 5.10%.

Solution:

Given the 30/360 day-count convention, there are 134 days between the last coupon on 1 March 2015 and the settlement date on 14 July 2015 (120 days for full months of March, April, May, and June, plus 14 days in July). Therefore, the fraction of the coupon period that has gone by is assumed to be 134/180. At the beginning of the period, there are 11 periods to maturity.

I) Stated annual YTM of 4.75%

The price at the beginning of the period is 101.198 per 100 of par value.

$$PV = \frac{2.5}{1.02375} + \frac{2.5}{1.02375^2} + \dots + \frac{102.5}{1.02375^{11}} = 101.198$$

$$\text{The full price on 14 July is } PV(\text{full}) = 101.198 \times \left(1.02375^{\frac{134}{180}}\right) = 102.979$$

$$\text{The accrued interest is } AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$$

$$\text{The flat price is } PV(\text{flat}) = 102.979 - 1.861 = 101.118$$

II) Stated annual YTM of 5.00%

The price at the beginning of the period is 100.00 per 100 of par value.

$$PV = \frac{2.5}{1.025} + \frac{2.5}{1.025^2} + \dots + \frac{102.5}{1.025^{11}} = 100.00$$

$$\text{The full price on 14 July is } PV(\text{full}) = 100.00 \times \left(1.025^{\frac{134}{180}}\right) = 101.855$$

$$\text{The accrued interest is } AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$$

$$\text{The flat price is } PV(\text{flat}) = 101.855 - 1.861 = 99.9942$$

III) Stated annual YTM of 5.10%

The price at the beginning of the period is 99.5256 per 100 of par value.

$$PV = \frac{2.5}{1.0255} + \frac{2.5}{1.0255^2} + \dots + \frac{102.5}{1.0255^{11}} = 99.5256$$

$$\text{The full price on 14 July is } PV(\text{full}) = 99.5256 \times \left(1.0255^{\frac{134}{180}}\right) = 101.4088$$

The accrued interest is $AI = 2.5 \times \left(\frac{134}{180}\right) = 1.861$

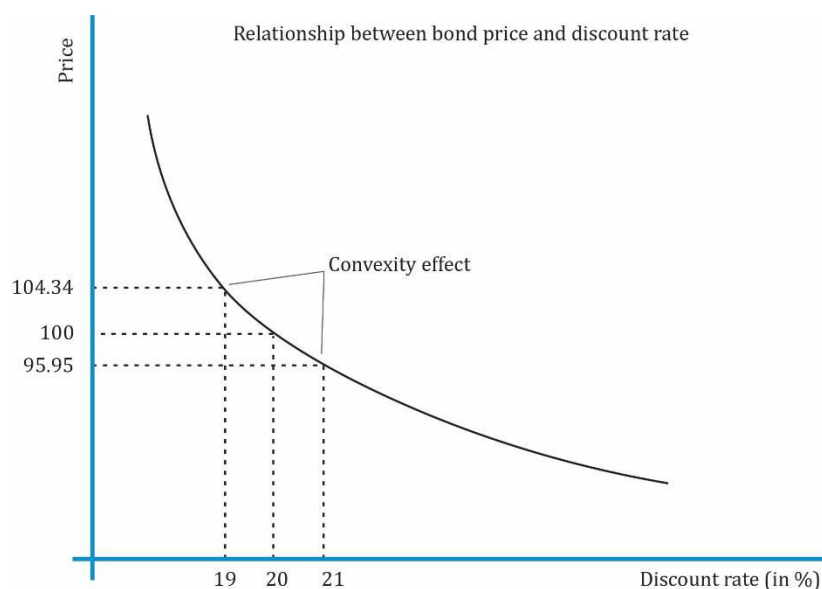
The flat price is $PV(\text{flat}) = 101.4088 - 1.861 = 99.5478$

The flat price of the bond is a little below par value, even though the coupon rate and the yield to maturity are equal, because the accrued interest does not take into account the time value of money. The accrued interest is the interest earned by the owner of the bond for the time between the last coupon payment and the settlement date. However, that interest income is not received until the next coupon date. The calculation of accrued interest in practice neglects the time value of money. Therefore, compared to theory, the reported accrued interest is a little “too high” and the flat price is a little “too low.”

3. Relationships Between Bond Prices and Bond Features

In this section, we will study the relationship between the bond price and market discount rate. The underlying principle is that a bond's price changes whenever there is a change in market discount rate.

Inverse Effect and Convexity Effect



The bond price is inversely related to the market discount rate. As you can see from the exhibit above, when the discount rate goes down, the price of the bond increases. This is called the inverse effect. Remember that the price-yield relationship of a bond is not a straight line. Also note that price changes are not linear for the same amount of change in discount rate. At an interest rate of 20%, the price of the bond is 100. If the interest rate decreases from 20% to 19%, the bond price increases by 4.34%. However, if the interest rate increases from 20% to 21%, the bond price decreases by only 4.05%. This difference in price change is called the convexity effect.

Coupon Effect

Consider three bonds (A, B, C) which have the same time to maturity but different coupon rates.

Bond	Coupon Rate	Price At 20%	Price At 19%	% Change	Price At 21%	% Change
A	10.00%	58.075	60.950	4.95%	55.405	-4.60%
B	20.00%	100.000	104.339	4.34%	95.946	-4.05%
C	30.00%	141.925	147.728	4.09%	136.487	-3.83%

Note that the bond with lowest coupon rate has the highest interest rate sensitivity. In other words, a 1% change in interest rates causes a greater % change in the price of bond A (10% coupon) relative to bonds B (20% coupon) and C (30% coupon). This is called the coupon effect.

Maturity Effect

For the same coupon rate, a longer-term bond has a greater percentage price change than a shorter-term bond when their market discount rates change by the same amount. This is called the maturity effect.

Summary of relationship between bond prices and discount rates	
Inverse effect	Bond price is inversely related to the discount rate.
Convexity effect	The percentage price change is more when discount rate goes down than when it goes up.
Coupon effect	For the same time to maturity and same change in discount rate the price of a low-coupon bond changes more than a high-coupon bond.
Maturity effect	For the same coupon rate and same change in discount rate, the price of a long-term bond changes more than a short-term bond.
Note: Low-coupon and Long-term bonds are more sensitive to changes in discount rates.	

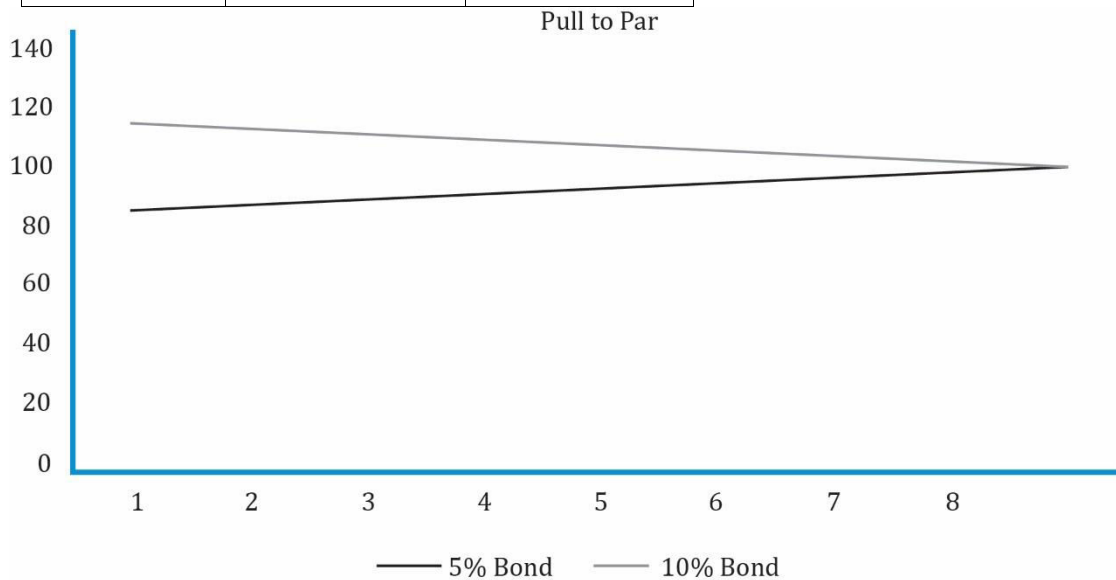
Constant-Yield Price Trajectory

Even if the yield stays constant, bond prices change and come closer to par as time passes and as they near maturity date. The 'constant-yield price trajectory' illustrates the change in a bond's price over time if yields remain constant. This trajectory shows the "pull to par" effect. For a premium bond, the price decreases over time to par and for a discount bond, the price increases over time to par.

The exhibit below shows the constant-yield price trajectories for 5% and 10% annual coupon payment, 8-year bonds. Both bonds have a market discount rate of 7.5%. The 5% bond's initial price is 85.3567 and the 10% bond's initial price is 114.6433.

Year	5% Bond	10% Bond
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1	85.3567	114.6433
2	86.7585	113.2415
3	88.2654	111.7346
4	89.8853	110.1147
5	91.6267	108.3733
6	93.4987	106.5013
7	95.5111	104.4889
8	97.6744	102.3256
	100.0000	100.0000



Instructor's Note

If the yield stays constant:

- A premium bond's price decreases to par value as its time to maturity approaches zero.
- A discount bond's price increases to par value as its time to maturity approaches zero.
- A par bond's value remains unchanged as it approaches maturity.

4. Matrix Pricing

Matrix Pricing Process

One can easily get information at what price a stock is trading. But it is not easy to determine the price of most bonds as they do not trade as often as equity. Matrix pricing is an estimation process to find the market price of a not-so-frequently traded bond based on the prices of comparable bonds with similar times to maturity, type of issuer, coupon rates, and credit quality. This is also applicable for bonds that have not been issued yet.

For example, consider a three-year, 4% semi-annual bond X that you are trying to value. Comparable bonds whose prices are known are shown in the matrix below. Based on the four bonds, we can determine the price of bond X.

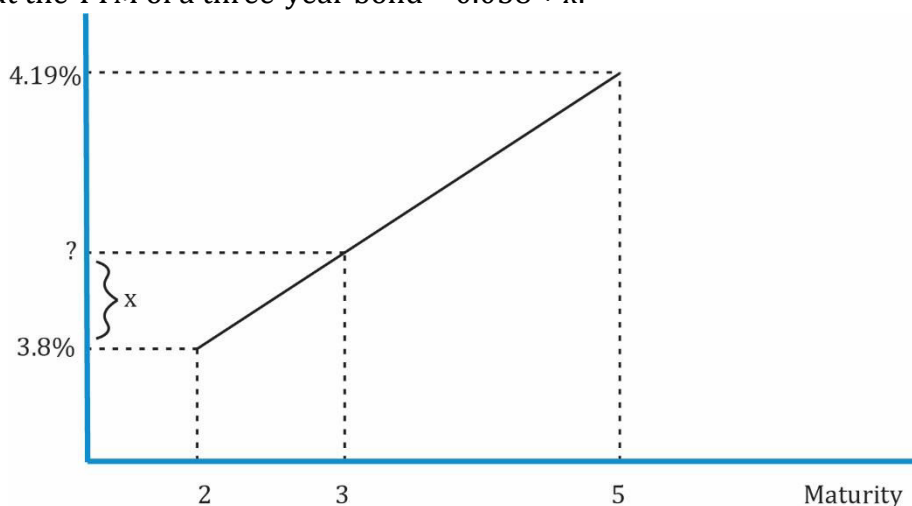
Matrix Pricing				
	2% coupon	3% coupon	4% coupon	5% coupon
Two years		98.5		102.25
		3.786%		3.821%
Three years			Bond X	
Four years				
Five years	90.25		99.125	
	4.181%		4.196%	

There are five steps to determine the value of a bond using the matrix pricing method:

1. Determine the YTM of comparable bonds, which is given to us already. If it were not given, you can calculate the YTM using the price and coupon. For example, YTM of a two-year, 3% coupon is: $N = 4$; $PMT = 1.5$; $FV = 100$; $PV = -98.5$; $CPT I/Y$. $I/Y = 3.786\%$.
2. Determine the average yield for two-year bonds:

$$\frac{0.03786 + 0.03821}{2} = 0.038035$$
3. Determine the average yield for five-year bonds:

$$\frac{0.04181 + 0.04196}{2} = 0.041885$$
4. Calculate the three-market discount rate using linear interpolation. We have the data (yield) for two-year and five-year bonds, but not a three-year bond. From the graph you can see that the YTM of a three-year bond = $0.038 + x$.



x can be calculated as $\frac{3-2}{5-2} \times (0.041885 - 0.038035) = 0.001283$. Discount rate for the three-year bond = $0.038035 + 0.001283 = 0.039318 = 3.93\%$.

5. Using 0.039318 as the discount rate, compute the price for the three-year bond.
 $N = 6$; $I/Y = 1.966$; $PMT = 2$; $FV = 100$; $CPT PV$. $PV = -100.1906$.

A few points to be noted on matrix pricing:

- Matrix pricing is used when underwriting new bonds to estimate the required yield spread over the benchmark rate.
- Benchmark rate is a market reference rate, or a government bond with the same maturity as the bond being priced.
- Assume the YTM for a new bond is calculated using the matrix pricing method as 2.2% and a comparable government bond has a yield of 2.0%. The difference between the yield on a new bond over its benchmark rate is called the **required yield spread** or spread over the benchmark. In this case, it is 0.2%.
- Yield spreads are always specified in basis points where 1 basis point is one-hundredth of a percentage point. In this case, it is 20 basis points.

Example

An analyst decides to value an illiquid 3-year, 3.75% annual coupon payment corporate bond. Given the following information and using matrix pricing, what is the estimated price of the illiquid bond? Additional information: 2-year, 4.75% annual coupon payment bond priced at 105.60 and 4-year, 3.5% annual coupon payment bond priced at 103.28.

Solution:

The required yield on the 2-year, 4.75% bond priced at 105.60 is 1.871%.

The required yield on the 4-year, 3.50% bond priced at 103.28 is 2.626%.

The estimated market discount rate for a 3-year bond having the same credit quality is the average of two required yields: $\frac{1.871\% + 2.626\%}{2} = 2.249\%$.

The estimated price of the illiquid 3-year, 3.75% annual coupon payment corporate bond is 104.3078 per 100 of par value.

$$\frac{3.75}{1.02249} + \frac{3.75}{1.02249^2} + \frac{103.75}{1.02249^3} = 104.3078$$

Summary

LO: Calculate a bond's price given a yield-to-maturity on or between coupon dates.

A bond's price is the present value of all future cash flows at the market discount rate, which is the rate of return required by investors given the risk of investment in the bond.

$$\text{PV of bond} = \frac{\text{PMT}}{(1+r)^1} + \frac{\text{PMT}}{(1+r)^2} + \cdots + \frac{\text{PMT}+\text{FV}}{(1+r)^N}$$

When a bond is between coupon dates, its price has two parts: flat price and accrued interest. The sum of the two parts is called the full price or dirty price.

Full price or dirty price = flat price + accrued interest

$$\text{Accrued interest} = \frac{t}{T} * \text{PMT}$$

Full price of a fixed-rate bond between coupon payments:

$$\text{PV}_{\text{full}} = \text{PV} * (1 + r)^{\frac{t}{T}}$$

LO: Identify the relationships among a bond's price, coupon rate, maturity, and yield-to-maturity.

Inverse effect	Bond price is inversely related to discount rate.
Convexity effect	For the same coupon/maturity bond: the percentage price change is more when discount rate goes down than when it goes up.
Coupon effect	For the same time to maturity and same change in discount rate: prices of low-coupon bonds change more than a high-coupon bond.
Maturity effect	For the same coupon rate and same change in discount rate: prices of long-term bonds change more than short-term bonds.
Note: Low-coupon and Long-term bonds are more sensitive to discount rates.	

LO: Describe matrix pricing.

Matrix pricing is an estimation process to find the market price of a not-so-frequently traded bond based on the prices of comparable bonds with similar times to maturity, type of issuer, coupon rates, and credit quality.

The steps involved in matrix pricing are:

- Determine the YTM of comparable bonds (which is usually given in the question).
- Determine the average yields of different comparable bonds.
- Calculate the market discount rate using linear interpolation.
- Using the discount rate, compute the price of the bond.